

Mobile Gait Analysis

Owen Agius
Department of Artificial Intelligence
Faculty of Information and Communication Technology
University of Malta
owen.agius.19@um.edu.mt

ABSTRACT

The scientific study of walking patterns is known as gait analysis. Typically, expensive and time-consuming marker-based technologies are employed in conjunction with several infrared cameras to generate kinematic data that concisely quantifies a person's walking behavior. Specialists can better comprehend an individual's walking behavior and gait using the kinematic data to detect gait-related disorders. The goal of this study is to create a mobile application which provides an automated alternative to gait analysis approaches based on markers.

Keywords

Mobile Gait Analysis, Deep Learning, Pose Estimation, Computer Vision, CNNs, GANs

1 Introduction

1.1 Motivation

A tremendous amount of technological progress has been made in recent years. Automation has been effectively adopted for various practical scenarios thanks to recent breakthroughs, minimizing the need for redundant user contact and making a range of tasks easier. The clinical task of gait analysis was used in this study. The possibility of automating the analysis is studied. The term "gait" refers to a person's walking style, which specialists can obtain insight into by analyzing with motion capture technology for the diagnosis of gait-related disorders. The most recent technical breakthroughs were investigated for gait analysis, to decrease user needs and the need for specialized equipment alongside high-priced hardware and software.

With permission, research was conducted at the bio-mechanics lab at the Mater Dei Hospital, guidance was provided by personnel with expertise in gait analysis and the equipment in the laboratory. The medical field presently analyzes gaits utilizing marker-based motion capture, in which anatomical markers are connected to a person and their movement is

documented during walking using infrared cameras. Current marker-based approaches include a lot of redundant stages that use too much time and exert too much effort.

Marker preparation and gait cycle selection are two of the most time-consuming phases in marker-based approaches. Marker attachment also necessitates that patients wear as little clothes as possible such that markers may more accurately estimate joint locations with less extraneous movement. Minimal clothing may place the patient in an uncomfortable position, causing their typical walking cycle to be disrupted. Marker-based approaches are also rather costly to set up, rendering them less accessible to clinics that may benefit from them. On the flip side, marker-less motion capture tends to provide more accessibility to automated implementations but have not yet been proven to perform to a high enough standard to be used in medical scenarios. Despite this, more marker-less methods have piqued the interest of a variety of research communities in the recent decade, which is expected given the growing demand for labor-saving motion capture. Visual hull generation (Mantoan et al., 2012; ; Bouwmans et al., 2019) was one of the contributions to marker-less motion capture, alongside, model development (Ceseracciu et al., 2014), and model validation (Ceseracciu et al., 2014) and model matching (Mundermann et al., 2007). More recently, automation for gait analysis has been explored by (Sammut-Bonnici, 2020) where both support vector machines (SVMs) and convolutional neural networks (CNNs) were used to classify abnormal or normal gait cycles by checking for exaggerated knee bending and too wide or too narrow step displacement. Still, as of late, the most important contribution has been deep learning-based since implementations based on deep learning have become more accurate, easy to acquire, and use in terms of accuracy, acquisition, and simplicity of use.

1.2 Goals and Objectives

The aim of this Final Year Project is to develop a mobile based automated substitute to marker-based gait analysis that requires less financial and computational expense and expertise. The alternative may serve as a baseline for future implementations that are cheaper, user-friendly and accessible to an ordinary smartphone. Through marker-less real-time motion capture, the proposed solution aims to achieve automation of gait analysis on a smartphone with respect to two main objectives:

1.2.1 How will abnormal gait cycles be classified

The first step would be pre-processing the captured datasets into a uniform format. The pre-processed videos would be used to create a generative adversarial network (GAN) to detect any anomalies in the gait cycle. Mobile Gait Analysis would be carried out by inputting the recorded walk from the smartphone into the GAN and checking for any anomalies.

1.2.2 How will the application be developed

A user-friendly interface will be developed where any necessary details from the user will be asked for upon account creation, such as age, gender, etc... The application would implement an interface for real-time motion capture in order to accommodate for mobile gait classification. Testing for the application would be carried out by participants being recorded in succession with the proposed mobile-based method and a sanctioned marker-based method, so that the resulting kinematics can be reviewed and evaluated by comparison.

2 Literature Review: Previous and Current Studies

This chapter goes over any previous or currently ongoing studies concerning pose-estimation, convolutional neural networks, automation for pose estimation and applications for pose estimation on mobile phones. These four areas of the literature review will provide a better understanding moving forward in building a mobile application for automated gait analysis

2.1 Pose Estimation

Pose estimation proposes the task of using a machine learning model in order to predict the pose of a person via a given image or video by estimating the structural configuration of key body joints (Piemontese, 2019). Pose estimation presents the tracking of an animate or inanimate object in 3-dimensional space at a granular level. Additionally, alongside human activity recognition, pose estimation opens up the opportunity for applications in augmented reality, animation, robotics and more relevant to our application, gait analysis. Gait analysis is a tool in which the walking and running cycles are assessed in order to attempt to identify any biomechanical abnormalities in the gait cycle. Gait analysis necessitates a high level of precision and accuracy in the pose estimation application in order for the gaits to be collected and plotted with a succession of poses that are consistent with each other. A pictorial structures approach and a more recently developed convolutional neural network approach are the two basic approaches to pose estimation.

2.2 Deep Learning

Convolutional neural networks were rendered much more popular when AlexNet (Krizhevsky et al., 2012) was developed for the ImageNet Large Scale Visual Recognition Challenge. AlexNet was a breakthrough in the development of CNN architecture due to the use of the Graphic Processing Unit for a fast training of the convolutional neural network, allowing 5 convolutional layers with 60 million parameters and 650,000 neurons to be able to run much quicker than before. Furthermore, AlexNet moved away from the use of the hyperbolic tangent for non-linear activation. Instead, AlexNet demonstrated the use of Rectified

Linear Units (ReLU) in order to reduce parameter complexity and dropout to effectively prevent overfitting. Although, this implementation adapts CNNs to perform better with a greater number of parameters and rising it to a new generation of computer vision research, AlexNet uses previous research (LeCun, Boser, et al., 1990). In the late 20th century (LeCun, Boser, et al., 1990; LeCun and Bengio, 1998) used previous research by (Fukushima, 1980) as a stepping stone, by suggesting the application of backpropagation to neural networks with convolutional layers for supervised machine learning algorithms. Convolutional layers can model data that has a specific number of dimensions. Layers of two dimensions are more commonly used on photographs, whereas one-dimensional layers are more commonly used on time series data. A multitude of channels make up the input data. The 2D input for colored images is made up of three channels, one for each of the red, green, and blue channels, but grayscale images just have one layer that stores the pixel intensity. The utilization of a GPU is especially important when dealing with RGB images since the number of convolutional operations required increases as the number of channels in the input increases. The network is created since each output neuron is connected to every other neuron from the input, forming fully connected layers and hence a network. However, if every output neuron is connected to every input neuron, this would generate an ineffective space since a connection for every combination of pixels would result in an extremely large parameter space. To solve this issue, weights were added to each neuron to suggest a healthier and more sensible connection between layers. Typically, a fully connected layer combines the outputs of the convolutional layers into a single 1D vector. Resultantly, the convolutional neural network to learn features within subdivisions of the image which proves to be fruitful in applications relating to image classification, such as pose estimation or object detection. A convolutional layer is made different from any other layer due to the implementation of a kernel applied to input artificial neurons in strides (Ciresan et al., 2011). Through these strides, the layer applies convolution to the input artificial neurons in groups relative to the size of the stride. Usually, a stride of tempo 1 is used, where after a group of neurons is convolved, the next group is selected to undergo convolution with a displacement size of 1. When the size of the kernel is greater than the stride, weight sharing takes place where some of the input artificial neurons in one group are also used in a neighbouring group. This proves to be fruitful in implementations consisting of pose estimation or object detection where these scenarios have features which are invariant in the image.

2.3 Automation for Pose Estimation

2.3.1 Pictorial Structures

Feature detection from images gained popularity in the medical field through a development in an object detection application aimed primarily on faces (Fischler and Elschlager, 1973). Furthermore, object detection gained traction for pose estimation later on when a new application was implemented which applied a set of descriptors referred to as parts to describe facial features such as the nose, mouth, eyes, etc... (Felzenszwalb and Huttenlocher, 2005). Detectable parts were arranged into a deformable configuration through "spring-like pairwise connections" (Felzenszwalb and Huttenlocher, 2005). Object detection on facial features can

be rather redundant and irrelevant for an implementation of pose estimation, therefore, rather than detecting features on the nose or mouth, body components such as the arms or the legs are connected through a series of “spring-like pairwise connections” in order to denote the body articulations, or more commonly known as the joints. Moving forward, a pictorial graph was plotted mimicking an undirected graph whose structure is determined by the categorical structure of the part locations. The locations consist of a combination of the part orientation, position and foreshortening. Further on, pictorial structures were outperformed through the introduction of Convolutional Neural Networks. Pictorial structures were observed to perform relatively poorly when considering multiple occluded joints or a pose estimation applied to a complex pose. The inefficiency became more prevalent due to the application of pictorial structures itself. The algorithm of pose estimation relies heavily on the location of local part descriptors rather than the global context of the image.

2.3.2 Introduction of Convolutional Neural Networks

DeepPose was the first study to implement convolutional neural networks over pose estimation (Toshev and Szegedy, 2014). DeepPose is based on classifying key-points in the pose which are generally but not limited to the major joints. The implementation is spread out over a convolutional neural network which accepts the whole image of the pose as an input and computes the features (joints) through each of its hidden layers. Many have taken inspiration from the DeepPose implementation and a wide variety of 2D pose estimation implementations have been researched and developed (Toshev and Szegedy, 2014). Any of these implementations can be viewed to be either in the top-down or bottom-up categories. The top-down approach entails only one pose being fed into the convolutional neural network at a time. With this in mind, detection must happen at an early stage which detects the human body and drawing a bounding box around the detected person. The bottom-up approach is less computationally expensive considering that detection occurs on the body parts first rather than the whole body. A link is plotted between the detected bodily features. The final stages are home to the construction of the final pose. Examples of bottom-up approaches are demonstrated but not limited by (Insafutdinov et al., 2016), (Li et al., 2019) and (Cao et al., 2017). More recently 3D approaches were explored. Although (Nibali et al., 2019) and (Moreno-Noguer, 2017) amongst others use this type of construction, these strategies are not as common as compared with 2D models due to a shortage in 3D datasets. Generally, 3D pose estimation implementations build on the two-dimensional implementations by firstly capturing the data and classifying in two dimensions and using the results captured in that dimension space, a pose in three-dimensional space is constructed. 3D constructions can be modelled using single monocular imagery such as (Nibali et al., 2019) and (Moreno-Noguer, 2017) which entail input from a single camera. Three-dimensional rotation and translation from a single 2D image or video are made possible by the technique presented in (Wattanacheep and Chitsobhuk, 2019). The pose is estimated directly from the 3D model pointed and the 2D image points where errors are corrected iteratively until a good pose-estimation is found from a single image (Wattanacheep and Chitsobhuk, 2019). But 3D con-

structions can also be modelled using multi-view inputs such as marker-based construction as more recently implemented by (Qiu et al., 2019) and (Iskakov et al., 2019). Although, multi-view models ensure a greater accuracy than monocular models, multi-view models require a much more expensive set up considering the equipment (cameras and laboratory) and the more time it takes to infer the pose from the markers. Thanks to the markers a skeleton can be formed. Recently any breakthroughs regarding pose-estimation are based on a skeleton-based representation of the human body such as (Presti and Cascia, 2016) and (Liu et al., 2017). This approach enables both 2D and 3D visual data where 3D data allows for additional depth information, whereas and more relevant to this study, 2D data allows a faster and simpler learning process. The skeleton-based approach has not yet been applied in gait analysis but it is far more popular in building human activity recognition models such as recent works by (Song, et al., 2017) which presents human activity recognition through the deep learning technique of building an RNN to build a spatio-temporal 3D skeleton model. Other work by (Ji et al., 2013) and (Zhao et al., 2017) promote the use of a 3D CNN rather than an RNN proposed by (Song, et al., 2017). Recent implementations of pose estimation are heavily reliant on deep learning algorithms such as recurrent neural networks (Song et al., 2017) and convolutional neural networks (Ji et al. 2013), (Zhao et al., 2017). Although, different neural network architectures are used for pose estimation, the approach consisting a CNN implementation is far more commonly used.

2.4 Mobile Pose Estimation

Now that pose estimation and deep learning have been covered, in this section, an added application for pose estimation is reviewed. The added implementation of a mobile application which estimates the pose in real-time was pioneered in the building of a mobile application which in real-time estimates the pose of the head of a person (Neto et al., 2012). Similar to previous pose estimation implementations, (Neto et al., 2012) promote the use of features or key points in their attempt to estimate the pose of a head. The model presented by (Neto et al., 2012) relies on the detection of the eyes and the nose through a 2D camera and convert each of the three key points into a 3D space using the roll, yaw and pitch equations proved by the great mathematician Leonhard Euler. These angles calculate the axes in which the object (in this case, the head) and its rotation relative to the Earth. More recently and slowly moving down the body, a model which attempts to estimate the pose of an upper body from a mobile application was built boasting a very high accuracy (Debnath et al., 2018). The pose estimation application for the upper body heavily relies on the implementation of MobileNets (Howard et al., 2017). Howard and his team present a class of efficient models for mobile or embedded vision applications labelled MobileNets. Improvements in pose estimation from mobile applications were made by the implementation of MobileNets by implementing the stacked hourglass network for pose estimation (Newell, Yang and Deng, 2016). This approach to building a CNN reduces the possibility of an overfitted architecture, a reduction of parameter size and an overall improvement in accuracy alongside an outperformance to MobileNets for pose estimation. Moreover, a combination of the skeletal-based approach and MobileNetsV2 to create a real-time 3D

human pose estimation model which works on a mobile device. The inspiration for MobileHumanPose was that currently 3D pose estimation models are not compatible with many mobile phones due to their low computational power, often reducing efficiency and accuracy (Choi et al., 2021). MobileHumanPose proposes a real-time 3D human pose estimation from a single-coloured image. Although this advancement is currently still being improved, the extra small model reduces inference time by 12.2ms on recently released mobile phones (Choi et al., 2021).

3 Methodology

The automation for gait analysis using a mobile phone is described in this chapter. A method for producing kinematics via posture estimation is provided, which automates the feature engineering element while also presenting a machine learning strategy for classifying aberrant and normal gait cycles. The development of a mobile application for gait analysis is also covered in this chapter.

3.1 Dataset Acquisition

In order to detect any irregularities in the gait of a person, the gait that has been pose estimated must be checked with a model. The model was created through multiple videos of normal gaits. The real-time video being captured is being matched with the model which points out any anomalies in the real-time gait.

The two datasets chosen being the OU-ISIR Biometric Database alongside the CASIA Gait Database were chosen due to both availability and ease-of-use, both databases are set-up in similar manners aiding the process of pre-processing ahead of gait analysis.

3.1.1 OU-ISIR Biometric Database

The OUISIR biometric database created by the Institute of Science and Industry at Osaka University contains 4,016 subjects aged 1 to 94 years. Data was collected from March 2009 as part of an information event in Japan. Approved informed consent was obtained from all subjects in this data set. The dataset consists of people walking on land surrounded by two cameras at 30 frames per second, 640 x 480 pixels.

3.1.2 CASIA Gait Database

Moving away from the structure of the OU-ISIR Biometric Database, the The Institute of Automation, Chinese Academy of Sciences (CASIA) opted to create three separate datasets being datasets A, B and C respectively.



Figure 1: Example of the 11 Different Captures in CASIA Dataset B (Source: Shuai Zheng, Junge Zhang, Kaiqi Huang, Ran He and Tieniu Tan, 2011)

"Dataset A was created in 2001 which contains 20 participants with 12 image sequences each. Dataset B is a large multiview gait database containing 124 participants in 11 different views. Three variations, namely view angle, clothing and carrying condition changes, are also captured. Finally, dataset C was collected by an infrared camera through 2005. The dataset contains 153 subjects and takes into account four walking conditions: normal walking, slow walking, fast walking, and normal walking with a bag." (Source: Shuai Zheng, Junge Zhang, Kaiqi Huang, Ran He and Tieniu Tan, 2011)

3.1.3 Pre-Processing of Datasets

Gait cycles are clustered in age groups and gender, in this study we will be working with the age range between 16 and 65, this is because gaits before 16 are vary too much due to puberty whereas gaits of people above 65 tend to suffer from motor issues that come with older age. Therefore, any data which is not in that specific age range will be filtered out. Since both datasets are labelled, it is easier to filter out the required data from the extra data. In the process of combining both of the datasets into one uniform dataset, the videos have been set into the same frame size 640 x 480. The final step of the pre-processing procedure is to extract the pose from the moving person, this decision was taken so that the machine learning classifier would not be prone to overfitting by searching for any extraneous details.

3.2 Pose Estimation

In a gait cycle, each pose is constructed by multiple keypoints, where each keypoint represents a body part location within the frame. The pose estimator which was implemented predicts 33 keypoints (as shown in Figures 2 and 3) where 3 of which are more relevant to gait analysis as they correspond to the hip, ankle and knee joints.

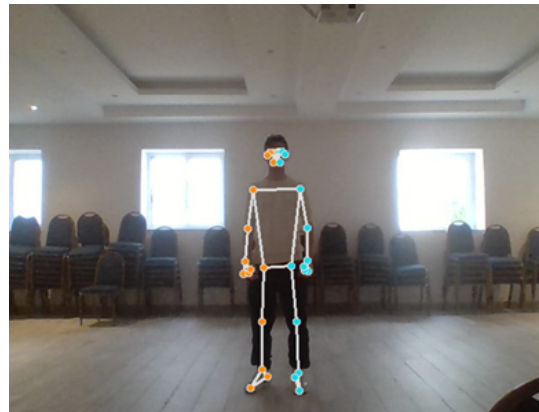


Figure 2: MediaPipe Representation of Pose Estimation. (Source: Owen Agius)

The technique used for the implementation of Pose Estimation is MediaPipe a subsidiary framework developed for BlazePose by Google (Bazarevsky et. al., 2020). The BlazePose framework was chosen to be implemented for this final year project due to a higher accuracy from any other implementations. BlazePose boasts an average 0.5% higher accuracy than AlphaPose (Fang et. al., 2017) alongside being able to run smoothly on less computationally powerful systems.

3.2.1 BlazePose

The method employs a two-step detector-tracker machine learning process, which first locates the person/pose ROI (Region of Interest) within the frame. Using the ROI-cropped frame as input, the tracker then predicts the posture landmarks and segmentation mask within the ROI. The detector is only used as needed in the video use cases, that is, for the first frame and when the tracker can no longer detect the presence of a body posture in the preceding frame. For other frames, the pipeline simply calculates the ROI based on the pose landmarks from the previous frame.

The pipeline is implemented as a MediaPipe graph that renders using a specific posture renderer sub-graph and uses a pose landmark sub-graph from the pose landmark module. A pose detection sub-graph from the pose detection module is used internally by the pose landmark. The detector is inspired from the BlazeFace model (Bazarevsky et. al., 2019), used in MediaPipe Face Detection, as a proxy for a person detector. The landmark model in MediaPipe Pose predicts the location of 33 pose landmarks. It predicts two more virtual keypoints that accurately characterize the human body's center, rotation, and scale as a circle.

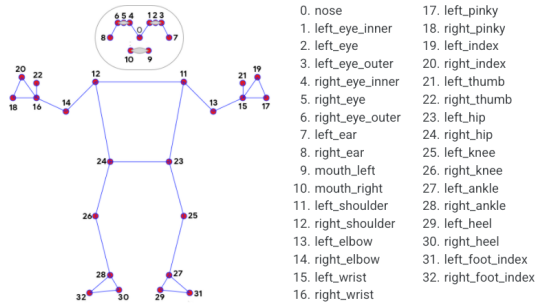


Figure 3: 33 Pose Landmarks (Source: (Bazarevsky et. al., 2020))

3.3 Proposed Continuation

3.3.1 Building the GAN

Although CNNs are far more frequently used when it comes to applications regarding pose estimation and gait analysis, an implementation of a GAN was opted to be used instead. More specifically, an implementation inspired by TadGAN (Geiger et. al., 2020). TadGAN present a new outlook to anomaly detection in which "several state-of-the-art techniques are built to compute reconstruction errors, as well as different approaches to combine reconstruction errors and critic outputs to compute anomaly scores" (Geiger et. al., 2020) alongside boasting a higher performance and accuracy than other previous anomaly detecting GAN implementations.

Both of the datasets mentioned in 3.1 will be pre-processed and then fed into the GAN. The GAN created would work in 6 stages.

1. Pre-processing the data.
2. Building a predictive machine learning (ML) model.
3. Make a prediction from the ML model for next the next movement.
4. Compute errors using the critic score functionality of

TadGAN.

5. Threshold the movement in the model alongside the movement recorded (as shown in Figure 4).

6. Detect anomalies in the pattern of movement.

Additionally, currently we are in contact with the University of South Florida for more gait data which would provide to be fruitful in creating a more diverse classifier.

The final step in the building of the GAN would be to upload the model to a web-based server to be able to be contacted by the mobile application via an API.

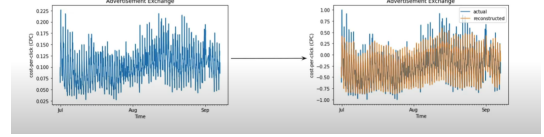


Figure 4: TadGAN Signal Thresholding (Source: (Geiger et. al., 2020))

3.3.2 Building the Application

The application would be built using the python framework 'KIVY' but if any issues arise in the implementation, a different route will be taken, being building a browser application which can be accessed through both computers and smartphones. The video captured from the application will be matched through the model outputted from the GAN (Mentioned in 3.3.1) through an API. The output from the application is ought to be similar to that of Run3D (as shown in Figure 5). This decision was taken because that representation is far more readable to the user, reducing the input of experts to explain the results of gait analysis.



Figure 5: Run3D Output (Source: Run3D (University of Oxford))

4 Evaluation Plan

The application built would be evaluated against a well-established maker-based gait analysis implementation. With permission, the bio-mechanics laboratory at Mater Dei Hospital are going to help out evaluating the Mobile Gait Analysis implementation by comparing the results, time efficiency and computational expense against the VICON system currently being used at their facilities.

The evaluation would first start off by testing the implementation on regular gait cycles, checking for any false negatives. The second step would be to test out on abnormal gait cycles mimicked by experts at the laboratory, so we know what output we are looking for. Both of the above steps would be carried out by select individuals working at the bio-mechanics laboratory. With permission from the individuals, the final step would be to test the proposed implementation on real abnormal data alongside the current implementation at the laboratory.

Errors and accuracies between both implementations would be compared by checking for the average and standard deviation of both front-view and side-view kinematics similar to the approach taken by (Sammut-Bonnici, 2020 (as shown in Figure 6)). Classifications outputted from the proposed system will be deemed correct or incorrect by the professionals working at the bio-mechanics laboratory.

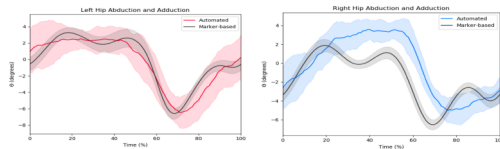


Figure 6: Error Metrics of Proposed System and VICON (Source: Sammut-Bonnici, 2020)

5 Conclusion

The anticipated result of this Final Year Project provide a working prototype of a mobile application which is able to classify in real-time any anomalies in the gait cycle of a person. The research presented in this report provides a better understanding to the different architectures used in previous studies and the new structure proposed in this final year project. The application of the machine learning technique to be able to classify any anomalies in the walking pattern is next on the agenda. The implementation and evaluation carried out will follow what has been mentioned in sections 3 and 4 unless any issues arise.

6 References

- [1] Mantoan, A., M. Cortesi, E. Ceseracciu, Z. Sawacha, S. Fantozzi, G. Gatta, and C. Cobelli (2012). "Markerless Gait Analysis: An Underwater Pilot Study". In: *Gait and Posture* 35.
- [2] Bouwmans, T., S. Javed, M. Sultana, and S. K. Jung (2019). "Deep Neural Network Concepts for Background Subtraction: A Systematic Review and Comparative Evaluation". In: *Neural Networks* 117, pp. 8–66.
- [3] Ceseracciu, E., Z. Sawacha, and C. Cobelli (2014). "Comparison of Markerless and MarkerBased Motion Capture Technologies through Simultaneous Data Collection during Gait: Proof of Concept". In: *PLOS ONE* 9.3, pp. 1–7.
- [4] Mundermann, L., S. Corazza, and T. P. Andriacchi (2007). "Accurately Measuring Human Movement using Articulated ICP with Soft-Joint Constraints and a Repository of Articulated Models". In: *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–6.
- [5] Piemontese, F. (2019). "Markerless Motion Analysis from Synchronized 2D Camera Views: A Convolutional Neural Network Approach". Thesis, University of Padova.
- [6] Fischler, M. A. and R. A. Elschlager (1973). "The Representation and Matching of Pictorial Structures". In: *IEEE Transactions on Computers* 1, pp. 67–92.
- [7] Felzenszwalb, P. F. and D. P. Huttenlocher (2005). "Pictorial Structures for Object Recognition". In: *International Journal of Computer Vision* 61.1, pp. 55–79.
- [8] Toshev, A. and C. Szegedy (2014). "DeepPose: Human Pose Estimation via Deep Neural Networks". In: *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1653, 1660.
- [9] Insafutdinov, E., L. Pishchulin, B. Andres, M. Andriluka, and B. Schiele (2016). "DeeperCut: A Deeper, Stronger, and Faster Multi-Person Pose Estimation Model". In: *European Conference on Computer Vision*. Springer, pp. 34–50.
- [10] Li, J., C. Wang, H. Zhu, Y. Mao, H. Fang, and C. Lu (2019). "CrowdPose: Efficient Crowded Scenes Pose Estimation and a New Benchmark". In: *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 10855–10864.
- [11] Cao, Z., T. Simon, S. Wei, and Y. Sheikh (2017). "Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields". In: *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7291–7299.
- [12] Nibali, A., Z. He, S. Morgan, and L. Prendergast (2019). "3D Human Pose estimation with 2D Marginal Heatmaps". In: *IEEE Conference on Applications of Computer Vision*, pp. 1477–1485.
- [13] Moreno-Noguer, F. (2017). "3D Human Pose Estimation from a Single Image via Distance Matrix Regression". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2823–2832.
- [14] Wattanacheep, Bhattarabhorn and Chitsobhuk, Orachat. (2019). "Prediction of 3D rotation and translation from 2D images. ICCCM 2019: Proceedings of the 2019 7th International Conference on Computer and Communications Management". 49–52.
- [15] Qiu, H., C. Wang, J. Wang, N. Wang, and W. Zeng (2019). "Cross View Fusion for 3D Human Pose Estimation". In: *IEEE International Conference on Computer Vision*, pp. 4342–4351.
- [16] Isakov, K., E. Burkov, V. Lempitsky, and Y. Malkov (2019). "Learnable Triangulation of Human Pose". In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 7718–7727.
- [17] Presti, L. L., and La Cascia, M. (2016). 3D skeleton-based human action classification: A survey. *Pattern Recognition*, 53, 130–147.
- [18] Chen, Y., Shen, C., Wei, X. S., Liu, L., and Yang, J. (2017). Adversarial posenet: A structure-aware convolutional network for human pose estimation. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 1212–1221).
- [19] Ince, O. F., Ince, I. F., Park, J. S., and Song, J. K. (2017, June). Gait analysis and identification based on joint information using RGB-depth camera. In *2017 14th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)* (pp. 561–563). IEEE.
- [20] Song, E., Soong, F. K., and Kang, H. G. (2017). Effective spectral and excitation modeling techniques for LSTM-RNN-based speech synthesis systems.

- IEEE/ACM Transactions on Audio, Speech, and Language Processing, 25(11), 2152-2161.
- [21] Zhao, W., Jiao, L., Ma, W., Zhao, J., Zhao, J., Liu, H., ... and Yang, S. (2017). Superpixel-based multiple local CNN for panchromatic and multispectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 55(7), 4141-4156.
 - [22] Ji S, Yang M, Yu K. 3D convolutional neural networks for human action recognition. *IEEE Trans Pattern Anal Mach Intell*. 2013.
 - [23] Krizhevsky, A., I. Sutskever, and G. Hinton (2012). "ImageNet Classification with Deep Convolutional Neural Networks". In: *Advances in Neural Information Processing Systems 25*, pp. 1097-1105.
 - [24] LeCun, Y., B. Boser, J. S. Denker, R. E. Howard, W. Hubbard, L. D. Jackel, and D. Henderson (1990). "Handwritten Digit Recognition with a Back-Propagation Network". In: *Advances in Neural Information Processing Systems 2*. San Francisco, CA, USA: Morgan Kaufmann Publishers, pp. 396-404.
 - [25] Fukushima, K. (1980). "Neocognitron: A Self-Organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by Shift in Position". In: *Biological Cybernetics* 36, pp. 193-202.
 - [26] LeCun, Y. and Y. Bengio (1998). "Convolutional Networks for Images, Speech, and Time Series". In: *The Handbook of Brain Theory and Neural Networks*. Cambridge, MA, USA: MIT Press, pp. 255-258.
 - [27] Ciresan, D., U. Meier, J. Masci, L. M. Gambardella, and J. Schmidhuber (2011). "Flexible, High Performance Convolutional Neural Networks for Image Classification." In: *International Joint Conference on Artificial Intelligence*, pp. 1237-1242.
 - [28] Neto, E.N., Duarte, R.M., Barreto, R.M., Magalhães, J.P., Bastos, C., Tsang, I., and Cavalcanti, G.D. (2014). Enhanced real-time head pose estimation system for mobile device. *Integr. Comput. Aided Eng.*, 21, 281-293.
 - [29] Debnath, B., O'Brien, M., Yamaguchi, M., and Behera, A. (2018, November). Adapting MobileNets for mobile based upper body pose estimation. In *2018 15th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)* (pp. 1-6). IEEE.
 - [30] Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., ... and Adam, H. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*.
 - [31] Newell, A., Yang, K., and Deng, J. (2016, October). Stacked hourglass networks for human pose estimation. In *European conference on computer vision* (pp. 483-499). Springer, Cham.
 - [32] Sangbum Choi, Seokeon Choi, Changick Kim; *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2021, pp. 2328-2338
 - [33] Sangbum Choi, Seokeon Choi, Changick Kim; *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2021, pp. 2328-2338
 - [34] S. Zheng, J. Zhang, K. Huang, R. He, and T. Tan, "Robust View Transformation Model for Gait Recognition," in *International Conference on Image Processing (ICIP)*, Brussels, Belgium, 2011
 - [35] Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., and Grundmann, M. (2020). BlazePose: On-device Real-time Body Pose tracking. *arXiv preprint arXiv:2006.10204*.
 - [36] Bazarevsky, V., Kartynnik, Y., Vakunov, A., Raveendran, K., and Grundmann, M. (2019). BlazeFace: Sub-millisecond neural face detection on mobile gpus. *arXiv preprint arXiv:1907.05047*.
 - [37] Thomas, D. M., Galbreath, D., Boucher, M., and Watts, K. (2020). Revisiting Leonardo da Vinci's Vitruvian Man Using Contemporary Measurements. *JAMA*, 323(22), 2342-2343.
 - [38] Geiger, A., Liu, D., Alnegheimish, S., Cuesta-Infante, A., and Veeramachaneni, K. (2020, December). TadGAN: Time series anomaly detection using generative adversarial networks. In *2020 IEEE International Conference on Big Data (Big Data)* (pp. 33-43). IEEE.
 - [39] Sammut-Bonnici, R. (2020, May). Automated Gait Analysis. University of Malta.
 - [40] Fang, Hao-Shu and Xie, Shuqin and Tai, Yu-Wing and Lu, Cewu. (2017) RMPE: Regional Multi-person Pose Estimation, *ICCV*.